

Expected free energy as an information constraint on the Bethe Lagrangian

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Abstract. Active inference selects actions by minimising an expected free energy functional of predicted futures. However, the expectation over yet-unobserved outcomes strips the functional of its Kullback-Leibler structure which hinders message passing treatments of inference procedures. Adding the expectation over future outcomes to the numerator restores the Kullback-Leibler structure but ultimately means that its stationary point no longer includes a mutual information term, which eliminates the agent’s epistemic drive. We propose an alternative formulation based on a Bethe free energy functional, fully supporting inference by message passing. The epistemic drive is maintained by imposing an information constraint, next to normalisation, marginalisation and form constraints, insisting that the mutual information between future observations, states and parameters given actions must be at least as large as the entropy of the goal prior. For a specific value of the corresponding Lagrange multiplier, the stationary point of this constrained Bethe Lagrangian recovers the EFE solution. We show that, as the information demand is varied, the solved Lagrange multiplier moves through its inactive, interior, and saturated regimes. In the inactive regime the agent’s epistemic drive switches off entirely, while in the saturated regime it is maximal. We test the performance of the constrained Bethe agent in an experiment with information gathering on a gridworld and one in which salience and novelty have to be balanced.

Keywords: Bethe free energy · Expected free energy · Constraints · Active inference · Message passing · Planning · Information Theory

1 Introduction

Active inference frames adaptive decision-making under uncertainty as the minimisation of an expected free energy functional (EFE), that, unlike reinforcement learning, naturally balances exploration and exploitation [2,14,10]. [That balance is what recommends EFE for control problems in which a system must be probed before it can be regulated \[5\].](#) Scaling active inference in structured generative models is typically done by casting inference as message passing on factor graphs [12,3], [which has been carried out for agents whose autoregressive likelihood parameters are learned online, with learning and planning realised as messages on a single graph \[6\].](#) The analogue of Friston’s free energy functional on

factor graphs is the Bethe free energy, which distributes energy terms to nodes, entropy terms to edges and avoids over-counting entropies around deterministic operation nodes [15,19,17]. The stationary points of the Bethe free energy functional correspond to messages that are passed between nodes and multiplied at edges to form marginal distributions. Which algorithm those messages implement is not fixed by the functional but by the local constraints imposed on it: sum-product, variational and expectation-propagation updates all arise as stationary points of the same Bethe Lagrangian under different constraints [20,17]. Constraint design is therefore where a new inference algorithm is specified, and it is where we intervene. The Bethe free energy has the structure of a Kullback-Leibler (KL) divergence, which allows its terms to be cleanly distributed over nodes and edges. EFE, however, does not have the structure of a KL divergence due to its expectation over yet-unobserved future outcomes. As such, its stationary points cannot be immediately distributed over its factor graph [7,8,18,13].

Efforts to address this discrepancy have been diverse. Schwöbel et al. propagated beliefs over future trajectories under a fixed expected-free-energy policy score [16]. Van de Laar et al. introduced a point-mass constraint on predicted outcomes that ensures that the constrained Bethe free energy functional experiences an epistemic drive [7]. Most recently, Nuijten et al. added epistemic priors to the generative model such that minimization of the Bethe free energy [13].

We follow Van de Laar et al.’s direction in that we formulate active inference as minimization of a constrained Bethe free energy functional, but impose an explicit information constraint instead. Specifically, we require that the mutual information between observations, states and parameters given actions, averaged over the variational posterior over actions, must be at least as large as the entropy of the goal prior distribution. Minimising the constrained Bethe Lagrangian under normalisation, marginalisation, form, and information constraints yields a stationary policy (14) that tilts the goal-weighted action prior by an exponential of the mutual information $I[x_t, \phi; y_t | u_\tau]$. Its weight γ_t is a Karush–Kuhn–Tucker multiplier solved from an information floor rather than tuned [1]. At $\gamma_t = 1$, EFE is recovered (c.f. Proposition 1). At $\gamma_t = 0$ the constraint is inactive and the agent experiences no epistemic drive. We test this constrained Bethe agent on a information-gathering task in a grid world environment with a goal cue and on a task where the agent must also learn a transition probability, comparing it to standard EFE and Q-learning agents [11].

2 Problem statement

Consider an agent that interacts with its environment through actions $u_t \in \mathcal{U}$ and observes outcomes $y_t \in \mathcal{Y}$, where $\mathcal{U}$ and $\mathcal{Y}$ are finite sets. It employs a probabilistic state-space model with discrete states $x_t \in \mathcal{X}$, transition parameters θ , and observation parameters ζ . At time step k , to plan for its next action, the agent unrolls the model forwards in time to a horizon of T time steps, collected as $\tau = (k+1, k+2, \dots, k+T)$, and attaches a goal prior $p_*(y_t)$ to every predicted

observation¹,

$$\tilde{p}(y_\tau, u_\tau, x_\tau, x_k, \theta, \zeta \mid \mathcal{D}_k) \propto \underbrace{p(x_k \mid \mathcal{D}_k)}_{f_k} \underbrace{p(\theta \mid \mathcal{D}_k)}_{f_\theta} \underbrace{p(\zeta \mid \mathcal{D}_k)}_{f_\zeta} \prod_{t \in \tau} \underbrace{p(x_t \mid x_{t-1}, u_t, \theta)}_{f_x} \underbrace{p(y_t \mid x_t, \zeta)}_{f_y} \underbrace{p(u_t)}_{f_u} \underbrace{p_*(y_t)}_{f_*}, \quad (1)$$

where $\mathcal{D}_k = \{y_{1:k}, u_{1:k}\}$. For brevity we collect the model parameters as $\phi = \{\theta, \zeta\}$ and the policy as $u_\tau = (u_{k+1}, \dots, u_{k+T})$ with prior $p(u_\tau) = \prod_{t \in \tau} p(u_t)$. The Forney-style factor graph of (1) is shown in Figure 1.

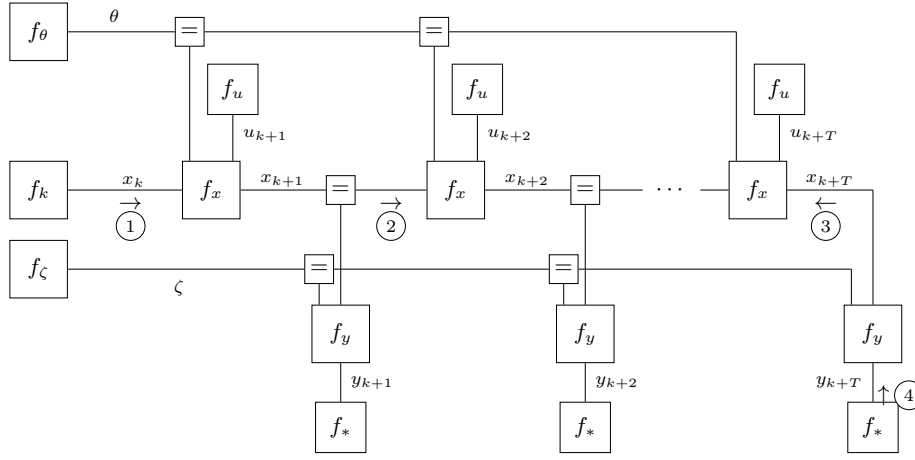


Fig. 1. Forney-style factor graph of the goal-augmented planning model in (1), unrolled over the horizon τ . Each transition factor (f_x) maps the previous state x_{t-1} , the control u_t and the transition parameters θ to x_t ; each emission factor (f_y) maps x_t and the emission parameters ζ to the predicted observation y_t , whose edge terminates in the goal-prior factor $f_* = p_*(y_t)$. Equality nodes ($=$) share θ and ζ across time and tie consecutive states together. Forward predictive messages (1, 2) and backward messages (3, 4) realise the forward-backward sweep of Sec. 3.

Expected free energy Expected free energy scores candidate actions by the surprise an agent expects to incur about its future observations under its preferences [2]. Writing $\tilde{q}(y_\tau, x_\tau, u_\tau, \zeta, \theta) = q(y_\tau \mid x_\tau, \zeta) q(x_\tau, u_\tau, \zeta, \theta)$ for the predictive extension of the variational posterior, the expected free energy functional is

$$\mathcal{G}[q] = \mathbb{E}_{\tilde{q}(y_\tau, x_\tau, u_\tau, \zeta, \theta)} \left[\ln \frac{q(x_\tau, u_\tau, \zeta, \theta)}{p(y_\tau, u_\tau, x_\tau, x_k, \theta, \zeta \mid \mathcal{D}_k)} \right]. \quad (2)$$

¹ Multiplying the forward model with the goal priors leaves the product unnormalized, hence the proportionality

For planning models with independent control priors $p(u_t)$, the optimal marginal variational factor for controls is $q(u_t) \propto p(u_t) \exp(-G(u_t))$ [7]. The expected free energy function $G(u_t)$ decomposes into risk and ambiguity terms;

$$G(u_t) = \mathbb{E}_{q(y_t, x_t, \theta, \zeta | u_t)} \left[\ln \frac{q(x_t, \theta, \zeta | u_t)}{p(y_t, x_t, \theta, \zeta | u_t, \mathcal{D}_k)} \right] \quad (3)$$

$$= \underbrace{D_{\text{KL}}[q(y_t | u_t) \| p_*(y_t)]}_{\text{risk}} + \underbrace{\mathbb{E}_{q(x_t, \theta, \zeta | u_t)} \left[H[p(y_t | x_t, \theta, \zeta, u_t)] \right]}_{\text{ambiguity}}. \quad (4)$$

Risk penalizes divergence between predicted outcomes and the goal prior, while ambiguity penalizes outcomes that are uninformative about the latent states and parameters. Thus, exploitation (low risk) and exploration (high information gain) emerge from a single objective. Re-arranging terms exposes [2]

$$I[x_t, \phi; y_t | u_t] = H[q(y_t | u_t)] - \mathbb{E}_{q(x_t, \phi | u_t)} [H[p(y_t | x_t, \phi)]] . \quad (5)$$

This is the predictive mutual information between the latents and the outcome, which allows us to express the negative expected free energy as a goal cross-entropy plus an information gain

$$-G(u_t) = \mathbb{E}_{q(y_t | u_t)} [\ln p_*(y_t)] + I[x_t, \phi; y_t | u_t] . \quad (6)$$

However, the expectation over the unobserved outcomes in (2) means the functional is no longer a Kullback–Leibler divergence, so the existing variational message updates technically do not hold [8,18,13]. From a graph perspective, this means that the resulting message at an f_x requires access to a node that it is not connected to, namely f_* .

The next section restores a message-passing treatment: the goal cross-entropy in (6) will emerge from the goal-prior factors of (1), and the information gain from a constraint on the Bethe Lagrangian.

3 Constrained Bethe free energy

The Bethe free energy of the planning graph collects one average-energy term per factor of (1), one entropy term for every variable and a term that prevents over-counting entropies based on the degree of each variable,

$$\begin{aligned} \mathcal{B}[q] = & \mathbb{E}_q \left[\ln \frac{q(x_k)}{p(x_k | \mathcal{D}_k)} \right] + \mathbb{E}_q \left[\ln \frac{q(\theta)}{p(\theta | \mathcal{D}_k)} \right] + \mathbb{E}_q \left[\ln \frac{q(\zeta)}{p(\zeta | \mathcal{D}_k)} \right] \\ & + \sum_{t \in \tau} \mathbb{E}_q \left[\ln \frac{q(x_t, x_{t-1}, u_t, \theta)}{p(x_t | x_{t-1}, u_t, \theta)} \right] + \sum_{t \in \tau} \mathbb{E}_q \left[\ln \frac{q(y_t, x_t, \zeta)}{p(y_t | x_t, \zeta)} \right] \\ & + \sum_{t \in \tau} \mathbb{E}_q \left[\ln \frac{q(u_t)}{p(u_t)} \right] + \sum_{t \in \tau} \mathbb{E}_q \left[\ln \frac{q(y_t)}{p_*(y_t)} \right] - \sum_{s \in \mathcal{S}} (d_s - 1) \mathbb{E}_q [\ln q(s)] , \end{aligned} \quad (7)$$

where $\mathcal{S} = \{\theta, \zeta, x_k\} \cup \{u_t, x_t, y_t\}_{t \in \tau}$ collects the variables shared across factors. The degrees are $d_\theta = d_\zeta = T + 1$ (one parameter prior plus T transitions or emissions), $d_{x_k} = d_{x_{k+T}} = 2$, $d_{x_t} = 3$ for $t \in \tau \setminus \{k + T\}$, and $d_{u_t} = 2$. The goal-prior factors raise the degree of the predicted observations to $d_{y_t} = 2$.

Constraints The Bethe free energy is minimized subject to four families of constraints on the variational factors. Firstly, normalization, requiring all variational factors to be valid probability distributions, and secondly, consistency across marginalisation for every cluster–edge incidence (a, i) in the planning graph:

$$\sum_{s_i} q_i(s_i) = 1, \quad \sum_{s_a \setminus s_i} q_a(s_a) = q_i(s_i). \quad (8)$$

Thirdly, we impose form constraints that pin the rolled out beliefs to the model’s own forward predictions.

Definition 1 (Form constraints). *The variational factors of the current state and parameter beliefs are constrained to match the generative model,*

$$q(x_k) = p(x_k | \mathcal{D}_k), \quad q(\theta) = p(\theta | \mathcal{D}_k) \quad \text{and} \quad q(\zeta) = p(\zeta | \mathcal{D}_k). \quad (9)$$

For $t \in \tau$, the variational factors of the state transition and likelihood match the generative model,

$$q(x_t | x_{t-1}, u_t, \theta) = p(x_t | x_{t-1}, u_t, \theta) \quad \text{and} \quad q(y_t | x_t, \zeta) = p(y_t | x_t, \zeta). \quad (10)$$

Lastly, a mutual information-based constraint targeting the epistemic term of expected free energy.

Definition 2 (Information constraint). *For each $t \in \tau$, the expected joint mutual information between the latent state and parameters on the one hand, and the predicted observation on the other, is at least the goal-prior entropy:*

$$\mathbb{E}_{q(u_\tau)} \left[I[x_t, \phi; y_t | u_\tau] \right] \geq H[p_*(y_t)]. \quad (11)$$

The information $I[x_t, \phi; y_t | u_\tau]$ is evaluated under the form constraints, so it is a fixed function of u_τ . It decomposes into state information gain $I[x_t; y_t | \phi, u_t]$ (salience), and parameter information gain $I[\phi; y_t | u_t]$ (novelty), thus addressing both epistemic drives.

The lower bound, i.e., the information floor $\beta_t \equiv H[p_*(y_t)]$, is based on the goal prior rather than set by hand. The agent must make the predicted outcome as informative about the latents as its preferences require. This has the added benefit of ensuring that the explore–exploit balance carries no free weight. The floor sums $H[p_*(y_t)]$ over the observation modalities, and which modalities enter it is a design choice.

Stationary points The form constraints ensure the only free belief left on the planning graph is the policy belief $q(u_\tau)$, and the Bethe free energy simplifies dramatically.

Lemma 1 (Bethe collapse). *Under the form constraints of Definition 1, the Bethe free energy (7) of the goal-augmented planning graph reduces exactly to*

$$\mathcal{B}[q] = D_{\text{KL}}[q(u_\tau) \parallel p(u_\tau)] - \sum_{t \in \tau} \mathbb{E}_{q(u_\tau)} \mathbb{E}_{q(y_t | u_\tau)} [\ln p_*(y_t)]. \quad (12)$$

Proof (sketch). Given u_τ , the constrained family equals the model’s own factorisation, so every transition and likelihood energy cancels against the corresponding entropy term. Only the policy prior and the goal factors survive. $\square$

The full accounting is given in Appendix A.1.

Introducing a multiplier λ for the normalization of $q(u_\tau)$ and $\gamma_t \geq 0$ for the information constraint at each $t \in \tau$, the constrained Lagrangian reads

$$\mathcal{L}[q] = \mathcal{B}[q] + \lambda \left(\sum_{u_\tau} q(u_\tau) - 1 \right) + \sum_{t \in \tau} \gamma_t \left(\beta_t - \mathbb{E}_{q(u_\tau)} \left[I[x_t, \phi; y_t | u_\tau] \right] \right), \quad (13)$$

where the information term is written in Karush–Kuhn–Tucker form [1]. The multiplier γ_t must be non-negative. Wherever the bound is slack, $\gamma_t = 0$. So, the multiplier switches on only when the agent would otherwise be insufficiently informative.

Theorem 1 (Stationary policy). *Every stationary point of the Lagrangian (13) subject to the form constraints (Definition 1) has policy belief*

$$q^*(u_\tau) \propto p(u_\tau) \exp \left(\sum_{t \in \tau} \mathbb{E}_{q(y_t | u_\tau)} [\ln p_*(y_t)] + \gamma_t I[x_t, \phi; y_t | u_\tau] \right), \quad (14)$$

with $\gamma_t \geq 0$, and control marginals $q^*(u_t) = \sum_{u_\tau \setminus u_t} q^*(u_\tau)$.

Proof (sketch). By Lemma 1 and the linearity of the constraint terms in $q(u_\tau)$, the variation of (13) with respect to $q(u_\tau)$ is

$$\frac{\delta \mathcal{L}}{\delta q(u_\tau)} = \ln q(u_\tau) + 1 - \ln p(u_\tau) + \lambda - \sum_{t \in \tau} \left(\mathbb{E}_{q(y_t | u_\tau)} [\ln p_*(y_t)] + \gamma_t I[x_t, \phi; y_t | u_\tau] \right). \quad (15)$$

Setting it to zero and normalising yields (14). $\gamma_t \geq 0$ holds by the Karush–Kuhn–Tucker conditions for (11). $\square$

The full proof is in Appendix A.2. The multipliers $\{\gamma_t\}$ are determined implicitly by the information constraints (11) and complementary slackness (Lemma 2 in Appendix A), while λ is fixed by normalization. [The implementation of section 3 solves them under a single floor for the whole horizon, which ties the \$\gamma_t\$ to a common value \$\gamma\$.](#) The stationary policy (14) modulates the action prior by the goal cross-entropy and by an exponential of the joint state-parameter–observation mutual information, an information-seeking bonus whose weight γ_t is solved rather than tuned. The relation to expected free energy is exact:

Proposition 1 (Equivalence to expected free energy). *If $\gamma_t = 1$ for all $t \in \tau$, then the stationary policy (14) coincides exactly with the expected-free-energy policy,*

$$q^*(u_\tau) = \propto p(u_\tau) \exp(-G(u_\tau)), \quad (16)$$

where $G(u_\tau) = \sum_{t \in \tau} G(u_t | u_\tau)$. *The condition is uniform across the horizon, so it is met by the shared multiplier of section 3 at $\gamma = 1$.*

The proof is in Appendix A.3.

Implementation The policy space is enumerated ($|\mathcal{U}|^T$ rollouts, vectorised over policies), and a single forward sweep through the transition and emission factors computes, for every u_τ and $t \in \tau$, the predictive beliefs, the goal cross-entropies $\mathbb{E}_{q(y_t|u_\tau)}[\ln p_*(y_t)]$, and the mutual informations $I_t(u_\tau)$ (salience plus novelty). Because the form constraints make every such functional linear in $q(u_\tau)$, no message iteration is required: the sweep converges in one pass, and the only iterative computation is the scalar dual.

For the dual we aggregate the per-step constraints (11) into one horizon constraint, $\mathbb{E}_{q(u_\tau)}[\sum_t I_t(u_\tau)] \geq \beta = \sum_t \beta_t$, with a single multiplier γ shared across the horizon. *This is a relaxation rather than a reformulation: meeting every per-step floor implies the summed one, but not conversely, so the agent is free to over-inform at one step and leave another slack. In the dual it amounts to restricting $(\gamma_1, \dots, \gamma_T)$ to the diagonal $\gamma_t \equiv \gamma$, and that diagonal is precisely where Proposition 1 lives, since expected free energy weights every step’s information by one. The shared multiplier is therefore the one-parameter family through the expected-free-energy point, and it replaces a T -dimensional root-find by the scalar one below. The two problems agree whenever the per-step multipliers coincide at the optimum, as they do when the floors and the attainable information are exchangeable across the horizon.* The expected information gain is monotonically non-decreasing in γ within the exponential family generated by (14) (Lemma 2 in Appendix A), so the solution falls into one of three cases by complementary slackness: an inactive constraint already met at $\gamma = 0$, a unique interior root of the binding condition (27) located by a Newton iteration with a bracketed-bisection fallback on $[0, \gamma_{\max}]$, or saturation at the bracket bound $\gamma_{\max}$ (10^3 in all experiments) when the demanded entropy exceeds the attainable information gain. Under the full-joint floor of our benchmarks the last case is realised: the floor outruns any rollout’s information, so γ saturates at $\gamma_{\max}$, which then plays the role of a maximal precision; restricting the floor to the preferred modality exercises the other two cases (Figure 3).

Acting requires a single decision rather than the full policy belief. The agent commits to the maximum a posteriori action of the stationary policy,

$$\hat{u}_{k+1} = \arg \max_{u_{k+1} \in \mathcal{U}} q^*(u_{k+1}), \quad (17)$$

with $q^*(u_{k+1})$ the first-step marginal of (14). It replans at every step.

4 Experiments

We run three experiments². The canonical T-maze empirically verifies the equivalence to EFE at $\gamma = 1$. The cue grid embeds the same epistemic structure in a navigation task where resolving the hidden reward condition requires a detour off the path to reward. The gate-cue task adds an unknown transition, so the agent must additionally value novelty.

All experiments run 200 trials from fixed seeds with $\gamma_{\max} = 10^3$. Goal priors are uniform on every observation modality except reward, where p_* places $(10^{-3}, 0.998, 10^{-3})$ on (null, reward, loss). In subsection 4.3 the Dirichlet prior over the unknown gate column has total concentration 4 (uniform mean) against 50 on the known transitions.

As baselines we use Q-MDP [11] and a standard expected-free-energy planner. Q-MDP is equipped with the true transition and emission, the regime in which the parameter posterior $q(\phi)$ collapses to a point mass and the novelty term in (14) vanishes. It acts by maximising the belief-averaged optimal Q-function of the underlying fully observable MDP over an HMM filtering belief. Because that Q-function assumes uncertainty resolves at no cost, Q-MDP assigns no value to information gathering, so the comparison isolates the information constraint (11) under a shared generative model. The EFE planner scores policies by $G(u_\tau)$ under the same goal prior and rollout, so it measures what the solved multiplier γ adds.

4.1 Canonical T-maze

The T-maze is a discrete benchmark in which information seeking is decisive [2]. The states are {center, cue, right arm, left arm} and the reward is $c \in \{\text{right}, \text{left}\}$, drawn uniformly at random across trials. The action $u_t \in \mathcal{U} = \{0, 1, 2, 3\}$ encodes the target location. The observation space $\mathcal{Y}$ is the triple (location, reward, cue). Entering the arm matching c emits a reward outcome with probability $p = 0.98$ and a loss outcome otherwise, while the worse arm flips these probabilities. Visiting the cue location deterministically reveals c , and elsewhere the cue observation is null. Each trial runs for $T = 3$ steps.

Results. Table 1 reports the full comparison, confirming that the information constraint (11) reproduces the cue-seeking that the unconstrained Q-MDP value cannot. Beyond trajectory agreement, we verify Proposition 1 directly: running the constrained Bethe agent with $\gamma = 1$ and a goal prior shared with the EFE agent, the two policy posteriors agree on every planning call of all 200 trials to machine precision, $\max_{u_\tau} |q^*(u_\tau) - q_{\text{EFE}}(u_\tau)| \leq 3 \cdot 10^{-16}$.

4.2 Information-gathering grid experiment

The hidden state factorises into a grid cell and a reward condition $c \in \{A, B\}$, fixed within a trial and uniform across trials, which selects the rewarding one

² Code available at <https://github.com/biaslab/IWAI2026-EFEasBetheConstraint>.

Table 1. Canonical T-maze over 200 trials ($T = 3$, $p = 0.98$; $\pm$ one standard error over trials). The constrained Bethe and standard-EFE agents behave identically; Q-MDP, blind to future information gain, gambles directly on an arm.

	Reach rate	Cue-visit rate	$\langle \text{time-to-target} \rangle$	$\langle \text{return} \rangle$
Q-MDP	$0.82 \pm .03$	$0.79 \pm .03$	$2.68 \pm .05$	$+1.22 \pm .06$
EFE	1.00	1.00	2.00	$+1.92 \pm .03$
CBFE	1.00	1.00	2.00	$+1.92 \pm .03$

of two absorbing goal cells. The action $u_t \in \mathcal{U} = \{\text{CCW}, \text{CW}, \text{FORWARD}\}$ drives deterministic egocentric navigation. The observation $y_t \in \mathcal{Y}$ is the quadruple (cell, heading, cue, reward): cell and heading are reported deterministically, the rewarding goal emits a reward outcome and the other a loss, and a cue cell at the end of a dead-end corridor reveals c when occupied. Unlike the T-maze, the cue is a multi-step detour away from both goals, so an agent that does not value information gain has no incentive to visit it. Each trial runs for at most $T_{\max} = 25$ steps with planning horizon $T = 6$, and the goal prior is peaked on the reward outcome as in the T-maze.

Results. The constrained Bethe agent visits the cue on every trial, detouring to resolve c before committing, and reaches the rewarding goal on $84.0 \pm 2.6\%$ of trials (Figure 2, left). The residual failures are horizon-induced dithering in the cue’s dead-end corridor, from which no 6-step rollout reaches a goal. EFE never visits the cue: the information gain of the detour is outweighed by the goal cross-entropy cost of postponing the goal, so it rushes to the nearest goal and gambles, reaching the rewarding one on $45.0 \pm 3.5\%$ of trials with negative mean return. Q-MDP likewise never values the off-path cue (a 1% incidental visit rate) and reaches the rewarding goal on only $36.5 \pm 3.4\%$ (Figure 2, right). Re-encoding the preferences in the canonical loss-averse log-preference form (EFE-LA in Table 2) rescues the EFE planner: a blind gamble then has negative value relative to waiting, and it visits the cue on every trial and reaches the rewarding goal on $99.5 \pm 0.5\%$, exceeding even the saturated constrained agent, which loses trials to post-cue dithering. The comparison points to a design trade-off: standard EFE obtains exploration through preference design, with the loss-aversion magnitudes implicitly setting the explore–exploit balance. The constrained agent demands it explicitly (Table 2).

Planning cost is dominated by the rollout shared with EFE (≈ 0.35 s per call). Solving the scalar dual adds under 3 ms.

The floor selects the regime. The benchmark above fixes the floor at the full-joint goal-prior entropy, which no rollout attains, so the dual saturates. The floor is restricted to the reward modality and varying its width $H[p_*]$ instead, exposes the full Karush–Kuhn–Tucker behaviour of Lemma 2 (Figure 3). For a representative planning state the solved multiplier is zero while the floor is

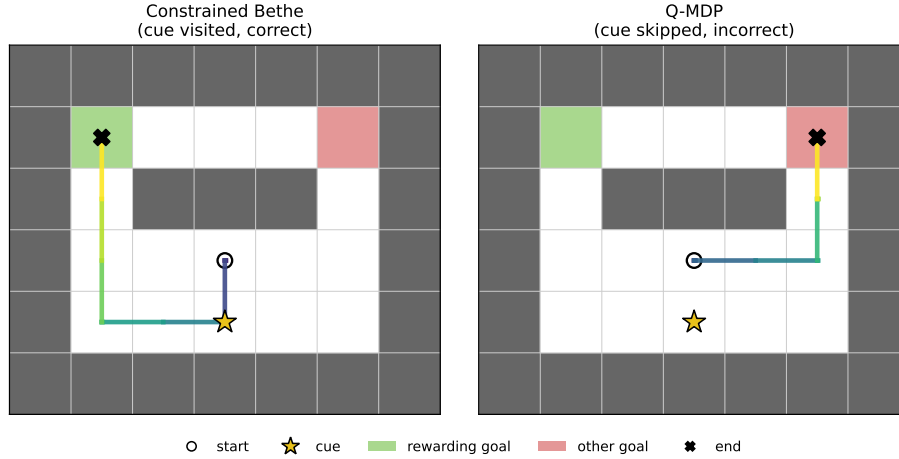


Fig. 2. A representative cue-grid trial. Left: the constrained Bethe agent detours to the off-path cue (gold star) to resolve the reward condition, then proceeds to the rewarding goal (green). Right: Q-MDP ignores the cue and commits to a goal directly, here the non-rewarding one (red). Walls are grey, the open circle is the start, and paths shade from light (early) to dark (late).

Table 2. Information-gathering grid over 200 trials ($T = 6$, $T_{\max} = 25$; $\pm$ one standard error). Reach rate is the fraction of trials reaching the rewarding goal. time-to-target is over the trials that reach a goal, which favours agents with lower reach rates. The unconditional return column is unaffected.

	Reach rate	Cue-visit rate	$\langle \text{time-to-target} \rangle$	$\langle \text{return} \rangle$
CBFE	$0.84 \pm .03$	1.00	12.49 ± 0.27	$+0.83 \pm .03$
EFE	$0.45 \pm .04$	0.00	6.00 ± 0.00	$-0.10 \pm .07$
EFE-LA	$1.00 \pm .01$	1.00	12.32 ± 0.18	$+1.00 \pm .01$
Q-MDP	$0.37 \pm .03$	$0.01 \pm .01$	13.08 ± 0.34	$+0.02 \pm .06$

already met, rises through a unique interior root (where it crosses the expected-free-energy value $\gamma = 1$) and saturates at $\gamma_{\max}$ only once the floor exceeds the most informative rollout (Figure 3a). Behaviourally the agent passes from a blind gamble (no cue visit) to maximal information seeking (cue every trial). The multiplier is thus genuinely solved from the information demand.

4.3 Saliency–novelty allocation under learning

The previous tasks isolate saliency: the dynamics are known and the only uncertainty is the hidden state. We now place the agent at a junction with two competing epistemic sources and let it learn. A hidden reward condition $c \in \{A, B\}$, uniform per trial, is revealed at a CUE (saliency). In a separate direction lies a GATE with an unknown transition. The agent holds a diffuse Dirichlet prior on the single column $\theta = p(x_t \mid x_{t-1} = \text{GATE}, u_t = \text{USE})$, whose true (fixed)

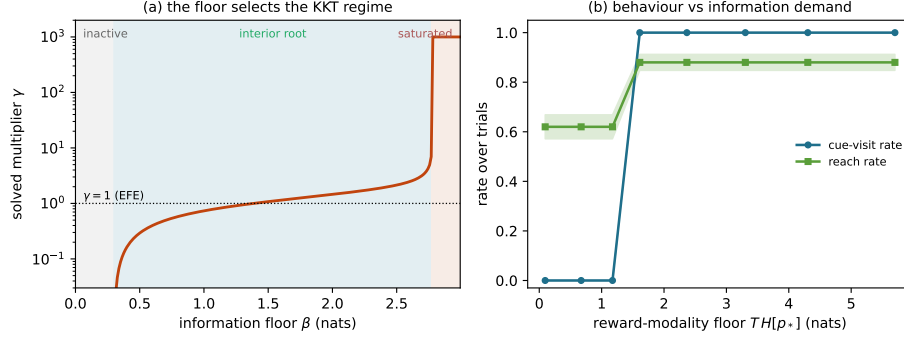


Fig. 3. The information floor selects the dual’s regime (cue grid). (a) Solved multiplier γ against the per-horizon floor β for a representative planning state: inactive at zero, a unique interior root crossing $\gamma = 1$, and saturation at $\gamma_{\max}$. (b) Cue-visit and rewarding-goal rates as the reward-modality floor $TH[p_*]$ is varied, rising from a blind gamble to maximal information seeking at the inactive–binding boundary of (a).

destination can only be learned by probing. Resolving c contributes salience $I[x_t; y_t | \phi, u_t]$, and probing the gate contributes novelty $I[\theta; y_t | u_t]$. The goals sit on two one-way three-step branches sharing only the start, so with horizon $T = 3$ the agent commits to one source per trial. Crucially, the Dirichlet posterior over θ persists across trials, so what the agent learns about the gate carries over. We set $\beta_t = H[p_*(y_t)]$, so the horizon floor is $\beta = \sum_{t \in \tau} \beta_t$, and run 200 trials.

Results. Early on, while θ is uncertain, novelty exceeds the cue salience and the single multiplier γ drives the agent down the gate branch. For the first 29 trials it probes the gate every time and, with c unresolved, gambles a goal (34% correct). As probing concentrates the Dirichlet posterior (gate-belief error falling from 0.83 to 0.18) novelty decays, and under the same constraint the agent’s effort shifts to the cue. It visits the cue on every subsequent trial, resolves c , and reaches the rewarding goal on 100% of them (Figure 4a,b).

Two controls isolate the mechanism: a salience-only agent given the true gate (point-mass θ , novelty zero) never probes and heads straight for the cue from the first trial. An agent whose Dirichlet posterior is reset each trial keeps probing indefinitely. Q-MDP, valuing neither source, performs at chance (49%).

A novelty-equipped standard EFE planner shows the same novelty to salience shift (Figure 4b) but switches earlier (last probe at trial 15; 95.5% correct overall vs. 90.5% for the saturated agent). This occurs because the unit multiplier weights information more weakly than the saturated dual. The mechanism is thus carried by the joint state–parameter information gain. in CBFE, a single solved multiplier produces an adaptive allocation that shifts from novelty- to salience-seeking as the agent learns its own dynamics.

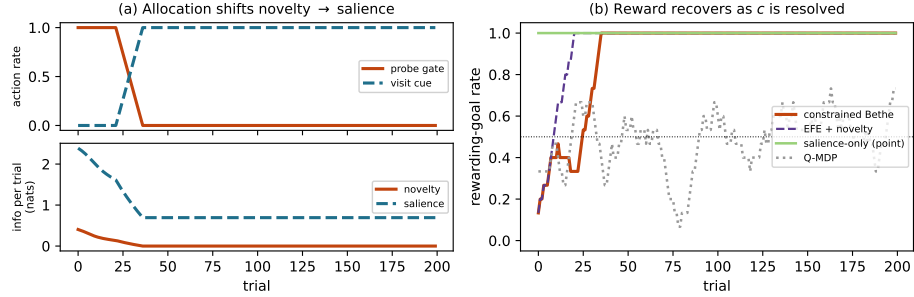


Fig. 4. Saliency–novelty allocation under learning. **(a)** The gate-probe rate falls and the cue-visit rate rises across trials (top), tracked by the per-trial novelty and saliency (bottom). **(b)** The rewarding-goal rate recovers to 1 as the agent stops gambling; the novelty-equipped EFE planner shifts earlier, the saliency-only agent is always correct, and Q-MDP is at chance.

5 Discussion

EFE-based planning can be cast as variational inference on a model augmented with preference and epistemic priors [18]. Epistemic priors realise information-seeking as a further model factor with a fixed weight, keeping the objective an ordinary variational free energy at the price of folding the epistemic drive into the agent’s model of the world. We realise it as an inequality constraint with a solved dual: the drive switches off, binds, or saturates by complementary slackness (Lemma 2). Constraints on the Lagrangian are powerful methods to enrich inference algorithms [17]. For example, Van de Laar et al. also casts planning as constrained Bethe minimisation through the same local constraint manipulations, but bounds the probability of violating an outcome specification – a safety requirement on the goal [9]. Both the information and the chance constraint are examples of encoding desired outcomes of the inference process, which, in our view, should be not be viewed as a priori knowledge.

Future work Fixing the floor at the goal-prior entropy means the explore–exploit balance is governed entirely by the goal prior. It is possible to add a slack variable to the constraint to decouple information-seeking from goal satisfaction. The form constraints are a restriction: open-loop rollouts do not condition on the observations a policy would in fact acquire, so multi-step information gains are overcounted relative to sophisticated, closed-loop planning [4]. Relaxing the form family toward conditional plans is a natural next step.

6 Conclusion

We posed action selection as constrained minimisation of a Bethe free energy on a goal-augmented planning graph. A single information constraint per step, a lower bound on the expected mutual information with a Karush–Kuhn–Tucker

multiplier that is solved rather than tuned, yields a one-parameter family of policies containing the expected-free-energy policy exactly at unit multiplier. The constrained Bethe agent matches a standard expected-free-energy planner to machine precision on the T-maze, detours to an off-path cue that an exploit-only Q-MDP baseline ignores, and reallocates its effort from novelty- to salience-seeking as it learns an unknown transition.

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A Appendix: proofs of theoretical results

Planning graph and constrained family. Let V denote the set of cluster (factor) nodes in the planning graph (Figure 1) and E the set of edges (shared variables). For (1) the cluster set is

$$V = \{x_k, \theta, \zeta\} \cup \{x_t, y_t, u_t, g_t\}_{t \in \tau}, \quad (18)$$

where each composite cluster is named, in sans serif, after the variable it emits – x_t for the transition, y_t for the emission, u_t for the control, and g_t for the goal cluster at time t – while the singleton prior clusters are named by their variables. The edge set is $E = \{\theta, \zeta\} \cup \{x_t\}_{t \in \{k\} \cup \tau} \cup \{u_t, y_t\}_{t \in \tau}$, with degrees as in Equation 7.

The form constraints of Definition 1 are local constraint manipulations in the sense of Şenöz et al. [17]. Fixing $q(x_k), q(\theta), q(\zeta)$ is a structured mean-field factorisation between the parameters and the trajectory with the parameter beliefs held at their past-graph values. Joining the control clusters $\{u_t\}_{t \in \tau}$ into a single policy cluster with belief $q(u_\tau)$ is a cluster (region) choice, and pinning the rollout conditionals is a form constraint on the conditional beliefs. Jointly they restrict the variational family to

$$q(y_\tau, x_{k:k+T}, u_\tau, \phi) = \quad (19)$$

$$q(u_\tau) p(x_k | \mathcal{D}_k) p(\theta | \mathcal{D}_k) p(\zeta | \mathcal{D}_k) \prod_{t \in \tau} p(x_t | x_{t-1}, u_t, \theta) p(y_t | x_t, \zeta),$$

in which the policy belief $q(u_\tau)$ is the only free factor, and all cluster and edge beliefs of (7) are the corresponding marginals of (19).

A.1 Proof of Lemma 1

Proof. Conditional on u_τ and with the parameter beliefs fixed, the family (19) factorises along a tree: the parameter-sharing equality chains of Figure 1 enter only through the fixed beliefs $q(\theta), q(\zeta)$, which severs every cycle. On a tree the Bethe entropy decomposition is exact [19], so the Bethe free energy (7), evaluated on the marginals of (19) with the controls collected in the policy cluster, equals the variational free energy of the joint,

$$\mathcal{B}[q] = \mathbb{E}_q \left[\ln \frac{q(y_\tau, x_{k:k+T}, u_\tau, \phi)}{\tilde{p}(y_\tau, u_\tau, x_\tau, x_k, \theta, \zeta \mid \mathcal{D}_k)} \right]. \quad (20)$$

Substituting (19) and the goal-augmented model (1), every factor of (19) cancels against its counterpart in $\tilde{p}$ – the priors $p(x_k \mid \mathcal{D}_k), p(\theta \mid \mathcal{D}_k), p(\zeta \mid \mathcal{D}_k)$ exactly, and the pinned transition and emission conditionals pairwise – leaving

$$\ln \frac{q(\cdot)}{\tilde{p}(\cdot)} = \ln \frac{q(u_\tau)}{p(u_\tau)} - \sum_{t \in \tau} \ln p_*(y_t), \quad (21)$$

up to the additive normalisation constant of $\tilde{p}$. Taking the expectation under (19) yields (12). $\square$

A.2 Proof of Theorem 1

Proof. The form constraints make $\mathbb{E}_{q(u_\tau)}[I_t] = \sum_{u_\tau} q(u_\tau) I[x_t, \phi; y_t \mid u_\tau]$ linear in $q(u_\tau)$ with fixed coefficients. By Lemma 1, the variation of the Lagrangian (13) with respect to $q(u_\tau)$ is

$$\frac{\delta \mathcal{L}}{\delta q(u_\tau)} = \ln q(u_\tau) + 1 - \ln p(u_\tau) + \lambda - \sum_{t \in \tau} \left(\mathbb{E}_{q(y_t \mid u_\tau)} [\ln p_*(y_t)] + \gamma_t I[x_t, \phi; y_t \mid u_\tau] \right). \quad (22)$$

Setting it to 0, solving for $\ln q(u_\tau)$ and fixing λ by normalisation gives (14). The control marginals follow by marginalising the policy cluster. The signs and feasibility of the duals $\gamma_t \geq 0$ are the Karush–Kuhn–Tucker conditions for the inequalities (11), resolved by Lemma 2 below. $\square$

A.3 Proof of Proposition 1

Proof. Per time step in the planning horizon, the identity (6) evaluated under the rollout reads $-G(u_t \mid u_\tau) = \mathbb{E}_{q(y_t \mid u_\tau)} [\ln p_*(y_t)] + I[x_t, \phi; y_t \mid u_\tau]$. Setting $\gamma_t = 1$ for all $t \in \tau$ in (14) therefore gives

$$q^*(u_\tau) \propto p(u_\tau) \exp \left(\sum_t -G(u_t \mid u_\tau) \right) \quad (23)$$

$$= p(u_\tau) \exp \left(-G(u_\tau) \right), \quad (24)$$

which is the solution under the EFE functional. $\square$

Determining the dual. As described in section 3, the implementation aggregates the per-step constraints into the single horizon constraint $\mathbb{E}_{q(u_\tau)} [\sum_{t \in \tau} I_t(u_\tau)] \geq \beta = \sum_{t \in \tau} \beta_t$ with one multiplier γ shared across the horizon, which preserves the Karush–Kuhn–Tucker structure with a single dual variable. The following lemma characterises its solution. The per-step system is the same object in T dimensions: writing $\ln Z(\gamma_1, \dots, \gamma_T) = \ln \sum_{u_\tau} \bar{p}(u_\tau) \exp(\sum_t \gamma_t I_t(u_\tau))$, the stationarity conditions of the per-step problem read $\nabla \ln Z = (\beta_1, \dots, \beta_T)$, with Jacobian $[\text{Cov}_{q^*}[I_t, I_s]]_{t,s} \succeq 0$. Hence $\ln Z$ is convex and $\nabla \ln Z$ is monotone, and the lemma below is its restriction to the diagonal $\gamma_t \equiv \gamma$, along which the Jacobian contracts to the scalar variance in (29).

Lemma 2 (Information multiplier). *Let the goal-tilted base measure be*

$$\bar{p}(u_\tau) \propto p(u_\tau) \exp \left(\sum_t \mathbb{E}_{q(y_t|u_\tau)} [\ln p_*(y_t)] \right), \quad (25)$$

and

$$q^*(u_\tau; \gamma) \propto \bar{p}(u_\tau) \exp \left(\gamma \sum_t I_t(u_\tau) \right) \quad (26)$$

be the measure that depends on the shared multiplier. The expected information gain $\mathbb{E}_{q^*(u_\tau; \gamma)} [\sum_t I_t]$ is non-decreasing in γ , rising from $\mathbb{E}_{\bar{p}} [\sum_t I_t]$ at $\gamma = 0$ to $\max_{u_\tau} \sum_t I_t$ as $\gamma \rightarrow \infty$.

The Karush–Kuhn–Tucker conditions admit exactly three cases:

1. if $\mathbb{E}_{\bar{p}} [\sum_t I_t] \geq \beta$ the constraint is slack and $\gamma = 0$;
2. if $\mathbb{E}_{\bar{p}} [\sum_t I_t] < \beta < \max_{u_\tau} \sum_t I_t$ the binding condition

$$\mathbb{E}_{q^*(u_\tau; \gamma)} \left[\sum_{t \in \tau} I_t(u_\tau) \right] = \beta \quad (27)$$

has a unique root $\gamma > 0$;

3. if $\beta \geq \max_{u_\tau} \sum_t I_t$ no policy can clear the floor and γ saturates at the bracket boundary $\gamma_{\max}$, concentrating the policy on the most informative roll-outs.

Proof. Write $S(u_\tau) = \sum_{t \in \tau} I_t(u_\tau)$ for the information a rollout supplies, and let

$$A(\gamma) = \ln \sum_{u_\tau} \bar{p}(u_\tau) \exp(\gamma S(u_\tau)) \quad (28)$$

be its log-partition function, so that $q^*(u_\tau; \gamma) = \bar{p}(u_\tau) \exp(\gamma S(u_\tau) - A(\gamma))$. The family $q^*(u_\tau; \gamma)$ is a one-parameter exponential family in the natural parameter γ with sufficient statistic S and base measure $\bar{p}$, so the derivatives of A return the cumulants of S ,

$$g(\gamma) \equiv \frac{dA}{d\gamma} = \mathbb{E}_{q^*(u_\tau; \gamma)} [S], \quad \frac{dg}{d\gamma} = \text{Var}_{q^*(u_\tau; \gamma)} [S] \geq 0. \quad (29)$$

The expected information gain is thus g , non-decreasing because A is convex. Its endpoints are the ones claimed. At $\gamma = 0$ the tilt is absent, leaving $g(0) = \mathbb{E}_{\bar{p}}[S]$. For the other endpoint let $S_{\max}$ be the largest value S takes on the support of $\bar{p}$ – which is the support of the control prior, the goal tilt in (25) being strictly positive – and let $\mathcal{U}^*$ collect the rollouts attaining it. For $u_\tau \notin \mathcal{U}^*$ and $u_\tau^* \in \mathcal{U}^*$,

$$\frac{q^*(u_\tau; \gamma)}{q^*(u_\tau^*; \gamma)} = \frac{\bar{p}(u_\tau)}{\bar{p}(u_\tau^*)} \exp(-\gamma[S_{\max} - S(u_\tau)]) \longrightarrow 0 \quad \text{as } \gamma \rightarrow \infty, \quad (30)$$

since the bracket is strictly positive. The policy space is finite, so the mass of the finitely many non-maximising rollouts vanishes, $q^*(\cdot; \gamma)$ concentrates on $\mathcal{U}^*$, and $g(\gamma) \rightarrow S_{\max}$.

Monotonicity alone does not give uniqueness; strictness does. By (29), $dg/d\gamma$ vanishes only if S is constant on the support of $\bar{p}$. In that degenerate case $\mathbb{E}_{\bar{p}}[S] = S_{\max}$, the interval $(\mathbb{E}_{\bar{p}}[S], S_{\max})$ is empty and case 2 cannot arise. Otherwise $dg/d\gamma > 0$ and g is a continuous, strictly increasing bijection from $[0, \infty)$ onto $[\mathbb{E}_{\bar{p}}[S], S_{\max})$. Note that $S_{\max}$ itself is approached but never attained at finite γ .

Besides stationarity, which produces the form $q^*(u_\tau; \gamma)$, the Karush–Kuhn–Tucker conditions are

$$g(\gamma) \geq \beta, \quad \gamma \geq 0, \quad \gamma[\beta - g(\gamma)] = 0, \quad (31)$$

that is, primal feasibility, dual feasibility and complementary slackness. The three cases are the trichotomy of β against the range of g .

1. If $\beta \leq g(0)$, then $\gamma = 0$ is already feasible and satisfies slackness. The constraint is inactive and the agent carries no epistemic drive.
2. If $g(0) < \beta < S_{\max}$, then $\gamma = 0$ is infeasible, so slackness forces $g(\gamma) = \beta$, which is (27). As β lies in the range of g and g is strictly increasing there, this root exists and is unique.
3. If $\beta \geq S_{\max} > g(0)$, then $g(\gamma) < \beta$ at every finite γ : no policy clears the floor, the feasible set is empty and no Karush–Kuhn–Tucker point exists. The multiplier grows without bound and the implementation truncates it at $\gamma_{\max}$, where q^* is concentrated on $\mathcal{U}^*$ – the maximal information-seeking limit.

□